Abstract

Urban green spaces are critical for maintaining biodiversity and supporting wildlife~~, especially in heavily urbanized environments.~~. In temperate regions, ~~winter brings~~resident birds face significant challenges ~~for resident bird species,~~ during winter, some of which ~~can~~may be ~~compounded~~exacerbated or, conversely, alleviated in urban ~~regions~~settings. To ~~better understand how birds~~ address those challenges or opportunities, even "resident" birds may complete very short-distance, within-city migrations to take advantage of seasonal habitats. To increase our understanding of non-breeding habitat use of resident birds in temperate urban ~~habitats during winter,~~ areas, we experimentally translocated Northern Cardinals (*Cardinalis cardinalis*) among sites spanning an urbanization gradient in Montreal, Canada~~, and used~~. We monitored individual movements throughout the non-breeding season using VHF telemetry to quantify ~~their~~ home range size, habitat selection across spatial scales, and landscape habitat associations. Cardinal populations on Christmas Bird Counts increased across the northeastern United States and southeastern Canada in the past 100 years ~~with most cardinals restricted to~~ , particularly in cities, ~~likely~~potentially due to ~~the presence of~~ anthropogenic food resources (e.g., bird feeders~~. Cardinals in~~). While cardinal core home range sizes during the ~~most urbanized study site had~~non-breeding period were similar across sites, the total home range was significantly larger ~~home ranges~~at urban sites compared to ~~birds wintering in the suburban and~~ peri-urban sites. ~~Habitat selection occurred~~Model fit was best at the smallest ~~scales measurable (25-50 m) across all sites (urban/suburban/peri-urban) implying very fine~~ scale ~~selection. This emphasises the importance of microhabitat features such as vegetation structure and bird feeder presence.~~across all site types. Habitat associations ~~varied~~differed across ~~the urban gradient~~site types, except for proximity to bird feeders, which was consistently related to habitat use by cardinals. ~~In less~~

~~urbanized areas, birds selected for vertically complex vegetation, likely because cardinals select shrubby, edge habitat. In more urbanized areas, birds were more likely to use areas with a greater density of roads, implying cardinals select the most residential areas. Cardinals exhibited behavioral plasticity in response to urbanization, suggesting that~~ In peri-urban sites, cardinal use was significantly greater at sites with lower vegetation and greater vertical complexity, suggesting a preference for heterogeneous or shrubby vegetation. We highlight the importance of fine-scale ecological understanding ~~can help~~to guide ~~locally tailored~~year-round green space management ~~aimed at supporting~~ for urban-adapted species ~~year-round~~.

Keywords: behavioral ~~plasticity~~flexibility, fine-scale movement, habitat association, home range size, manual VHF telemetry, non-breeding period, scale of habitat selection, urbanization gradient

Lay summary

- Urban green spaces can help wildlife survive in cities, but little is known about how birds use these spaces during the winter months in northern climates.
- We tracked Northern Cardinals in Montreal, Canada, using small radio transmitters to ~~follow where they move during winter.~~document movements and discover the habitats used
- ~~By tracking these birds, we quantified how much space they used and what kinds of places they favoured in areas ranging from greener neighbourhoods to denser parts of the city.~~
- Cardinals spending the winter in the most built-up areas used larger territories and ~~relied on resources found close by, such as vegetation features and human-provided food~~

~~sources like~~were most often in areas with bird feeders~~.~~ and lots of roads, such as residential areas

- ~~Overall, cardinals showed a strong ability to adjust to city life, suggesting that~~Cardinals show flexibility in how they use habitat depending on how urban it is, suggesting well-designed urban green spaces can support urban-adapted birds year-round.

~~**Introduction**~~

~~Urbanization is accelerating worldwide (Wang et al. 2012; Elmqvist et al. 2013; Simkin et al. 2022), altering landscapes and bringing novel challenges to wildlife including habitat fragmentation, habitat loss, and chronic pollution (Lowry et al. 2013; Seress and Liker 2015; Murray et al. 2019; Vaz et al. 2023). Future global urban expansion could lead to an additional 11-33 million hectares of habitat loss by 2100, with a disproportionately large increase in habitat fragmentation (Li et al. 2022).~~

~~This habitat loss and fragmentation can in turn reshape wildlife spatial ecology, including patterns of home range size and movement. A meta-analysis showed both birds and mammals had smaller home ranges in urban areas and that urbanization restricts spatial movements for most animals~~ **Introduction**

Urbanization is accelerating worldwide (Wang et al. 2012; Elmqvist et al. 2013; Simkin et al. 2022), leading to habitat fragmentation, habitat loss, and chronic pollution in and near urban landscapes (Lowry et al. 2013; Seress and Liker 2015; Murray et al. 2019; Vaz et al. 2023). Future global urban expansion could lead to an additional 11-33 million hectares of habitat loss by 2100, with a disproportionately large increase in habitat fragmentation (Li et al. 2022).

Urbanization in turn reshapes wildlife spatial ecology, habitat use, and may offer anthropogenic resources that species with behavioural plasticity may exploit for enhanced fitness opportunities. Urban landscapes, and particularly anthropogenic features such as buildings, roads, and other impervious surfaces, may restrict or alter animal movement or habitat use. This may in turn lead to different patterns of home range sizes and movement patterns for individuals of the same species within vs. outside of urban areas (O'Donnell and delBarco-Trillo 2020). During the breeding period, the communication distance of *Zonotrichia leucophrys* (White-crowned Sparrows) in the San-Francisco Bay was highly predictive of its territory size, highlighting that in the urban landscape, noise pollution may drive songbirds into smaller home ranges in the breeding season (Simberloff et al. 2023). Nonetheless, most research on urban birds occurs in the breeding season, and there is far less in winter, and those studies that do exist contrast with breeding season studies. This knowledge gap is important because birds often exhibit different behaviours outside of the breeding season. For example, *Cardinalis cardinalis* (Northern Cardinals) form winter flocks (while vigorously defending their territories against conspecifics during the breeding season) and alter their patterns of space use during the non-breeding season in Kentucky (Ritchison and Omer 1990). Consequently, factors influencing habitat use during winter may differ substantially from those operating during the breeding season.

Birds. An extensive meta-analysis found that several small bird and mammal species occupied smaller home ranges, with restricted spatial movement, in urban areas (O'Donnell and delBarco-Trillo 2020; but see opposite trend for predators: Riley et al. 2003; Thompson et al. 2021; Hindmarch et al. 2017). Smaller home ranges may be a result of locally abundant and reliable resources such as food or shelter, diminishing the need for extensive movement (Rutz 2006; Bino et al. 2010; Šálek et al. 2015; Reher et al. 2016; Mitsuhashi et al. 2018), or a high density of

conspecific competitors forcing limited territory use (Galeotti 1994; Boggie and Mannan 2014). Additional ecological drivers beyond resource use may influence home range sizes. For example, it was suggested that for *Zonotrichia leucophrys* (White-crowned Sparrows) in the San Francisco Bay area, limitations of communication distance in a noisy urban landscape may drive individuals into smaller home ranges during the breeding season (Simberloff et al. 2023). An individual's potential to respond to these urban pressures through behavioural adjustments~~,~~ ~~reflecting both~~ ultimately depends on the species' behavioural flexibility. Urban areas hold both novel challenges and opportunities, including increased predation pressure, ~~heightened disturbances~~higher disturbance levels, novel anthropogenic resources, and human-made structures. To compensate for disturbances from noise, light, and human activity, some ~~birds~~species may ~~change~~alter their nesting behaviour~~ for example by~~, including nesting higher in the canopy (e.g., *Pica pica* (Eurasian Magpie); Šálek et al. 2020) or using off-ground artificial surfaces ~~increasing~~, leading to increased fitness (e.g., *Junco hyemalis* (Dark-eyed Junco); Bressler et al., 2020). Urban habitats ~~can~~ also provide foraging opportunities which can be exploited by some species like *Falco peregrinus* (Peregrine Falcon~~). *F. peregrinus* was shown to use~~), which uses buildings as perches and nesting sites, capitalizing on disoriented migratory birds ~~wandering near~~attracted to lit glass structures (Kauffman et al. 2003; DeCandido and Allen 2006; Altwegg et al. 2014). ~~Although some species can adapt and thrive in urban areas, urban tolerance varies across species and across annual cycles, with urban tolerance peaking during the winter months (potentially due to the urban heat island effect) (Callaghan et al. 2021). Habitat use in the city can differ significantly from that in rural environments, even within a single species. These differences can reflect hierarchical habitat selection (Johnson 1980), as animals select habitat at multiple spatial scales – from geographic range (first order), to the home~~

range of an individual (second order), to the selection of specific patches for daily activities (third and fourth orders). In the urban context those decisions are influenced by novel pressures such as human disturbances, artificial structures, and altered resource availability. Quantifying habitat use by comparing used locations with what is available in the habitat can provide valuable insights into how wildlife responds to urbanization and identify habitat features that are more strongly associated with both occurrence and space use (Manly et al. 2002; Poirier et al. 2024).

Urban birds may outperform rural birds during winter, when natural resources are less available and weather is harsher. Cities can provide a more stable wintering environment for bird species that use feeders or that are generalists or omnivores (i.e., corvids), which exploit anthropogenic resources in winter (Jokimäki et al. 2022). In other cases, cities can serve as refuge during periods of extreme weather, as local colonization of cities by birds increased during the colder conditions due to cities acting as heat islands (Latimer and Zuckerberg 2021). Nonetheless, occupancy dynamics in temperate regions during winter are highly species-specific, with thermal tolerance playing an important role in determining which species would be able to persist in urban environments during this season. Moreover, energy demands for small resident passerines increase with latitude in winter, meaning that birds spending their winter in regions such as southern Quebec must consume significantly more energy to maintain their body temperatures than those wintering farther south – which is especially challenging given the scarcity of resources during the cold months (Fristoe 2015). Despite these physiological and environmental challenges, along with the potential added effects of urbanization, winter remains an underrepresented season in experimental, ecological research (Dinh et al. 2023). This gap in knowledge highlights the importance of investigating how birds are occupying urban habitats in

~~winter, particularly in temperate regions where winter survival is a major limiting factor for resident bird species.~~

~~To address this gap, we studied *Cardinalis cardinalis* (Northern Cardinal) across an urban gradient in one of Canada's largest cities, Montreal, Quebec. Cardinals are large songbirds (weighing 40 to 45 grams), which allows for deployment of radio tags capable of lasting almost the entirety of the non-breeding season. In the last 100 years, cardinals expanded their distribution range northwards, and have been anecdotally closely associated to urban and suburban landscapes in those areas (Beddall 1963; Smith et al. 2011; Robert et al. 2019). They are able to thrive and persist across a wide range of urbanization gradients, from suburban neighbourhoods to densely populated areas of the city. Cardinals are now resident in southern Quebec, a region of extreme cold (sometimes reaching -27 °C with wind chill; Environment and Climate Change Canada) and limited natural food sources during the winter months (Government of Canada 2004; Turcotte and Desrochers 2005). As part of a translocation experiment, we captured and released individuals at sites distributed along this urbanization gradient in Montreal to evaluated how they are using such habitat. We tracked weekly movements to assess how the birds adjust their movements and habitat use after being displaced into environments with varying levels of urbanization.~~

~~This study has four objectives: (i) investigate how cardinals' home range size varies across an urbanization gradient through the winter season, (ii) quantify the scale at which cardinals are associated with certain factors of their habitat, (iii) determine the relative strength of anthropogenic and natural vegetation to predict habitat use by cardinals during the non-breeding season, and (iv) document the spring movements from non-breeding to breeding grounds. We predicted that wintering cardinals require larger home ranges (both core and total) and select~~

~~habitats at larger scales when in dense urban locations due to food sources being more scattered (Havlíček et al. 2022). In addition, we predicted that locations with bird feeders, lower human population density and a complex vertical vegetation structure would be associated with a higher probability of cardinals. Finally, we predicted that cardinals would transition from their breeding sites in forested/canopy covered areas towards residential areas in the fall to take advantage of bird feeders during winter, reversing the movement in spring.~~

Similar to many wildlife studies (Marra et al. 2015), urban ecology research is strongly biased toward the breeding season. This is problematic because birds often exhibit greater urban tolerance and behavioural flexibility during the non-breeding season (Callaghan et al. 2021). Urban environments may also provide important winter refuges through predictable anthropogenic food sources and the urban heat island effect (Jokimäki et al. 2022; Latimer and Zuckerberg 2021). Winter occupancy dynamics are highly species-specific and influenced by thermal tolerance, particularly at northern latitudes where energetic demands increase substantially (Fristoe 2015). For example, *Cardinalis cardinalis* (Northern Cardinals) form winter flocks and alter their space use outside the breeding season (Ritchison and Omer 1990). Birds overwintering at northern latitudes, such as southern Quebec, are expected to spend considerably more energy maintaining body temperature than those wintering farther south, which is especially challenging given the scarcity of food resources during the cold months. Despite these challenges, winter remains underrepresented in ecological research (Dinh et al. 2023), highlighting the need to better understand how birds use urban habitats during this critical season.

To address the gap in knowledge about winter-resident urban birds, we studied *Cardinalis cardinalis* across an urban gradient in one of Canada's largest cities, Montreal, Quebec.

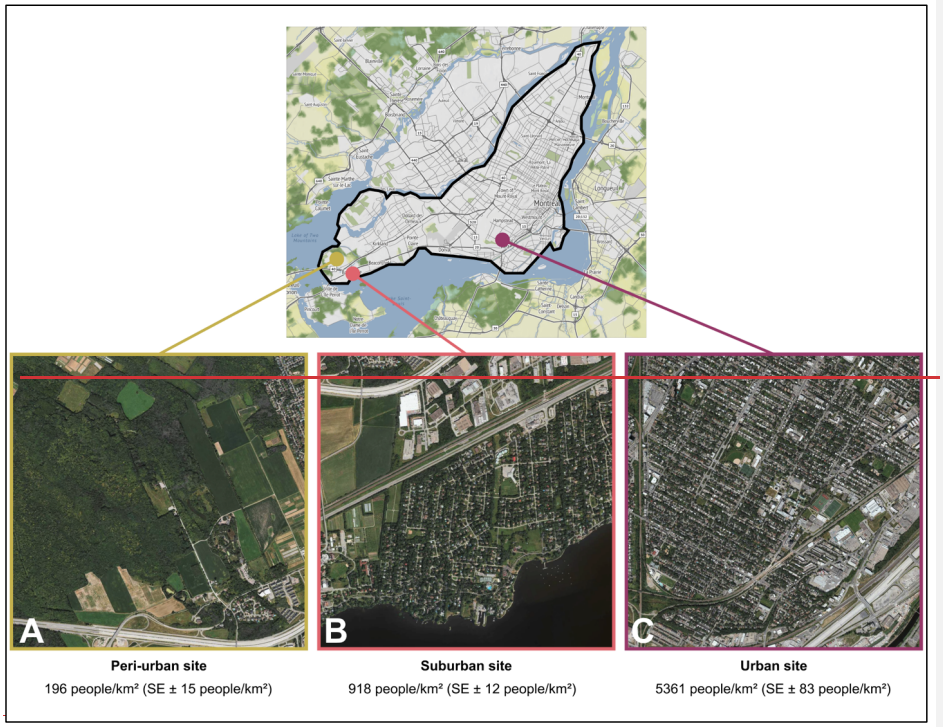
Cardinals are medium-sized songbirds (weighing 40 to 45 grams), capable of supporting radio tags lasting almost the entirety of the non-breeding season. Over the last 100 years, cardinals have expanded their distribution northward, especially in association with urban and suburban landscapes, and now occur in regions of extreme cold and limited winter resources, such as southern Quebec (Beddall 1963; Smith et al. 2011; Robert et al. 2019). As a translocation experiment, we captured and released individual cardinals at sites distributed along an urban gradient to broadly quantify habitat use and movement dynamics as a function of urbanization during the non-breeding season. Our study had four objectives: (i) to investigate cardinals' home range size as a function of urbanization, (ii) to quantify the spatial scale of habitat use associations, (iii) to determine possible predictors of cardinal habitat use from anthropogenic and natural habitat variables, and (iv) to qualitatively document the spring movements from non-breeding to breeding territories. We predicted that wintering cardinals require larger home ranges and select habitats at larger scales when in dense urban locations due to food sources being more scattered than in less urban areas (suburban/peri-urban; (Havlíček et al. 2022). We also predicted cardinal habitat use will be positively associated with food resources (bird feeders), preferred vegetation structure (heterogeneous vertical vegetation structure), and lower human population density. Finally, we predicted that cardinals would transition from their breeding sites in forested/canopy covered areas towards residential areas in the fall to take advantage of bird feeders during winter and reverse this localized migration in spring.

Materials and methods

Study locations

The island of Montreal is located in southern Quebec, Canada, with a population of ~2 million people ~~coupled with~~, including areas of high population density. The ~~city's~~local climate ~~is~~

~~comprised of~~comprises four distinct seasons with cold winters (temperatures can range between -23 ˚C and 4 ˚C during the month of January) and warm summers (with temperatures sometimes exceeding 30 ˚C; WeatherSpark). ~~This research was~~We conducted our research at three distinct study sites (Figure 1) ~~with varying~~reflecting different levels of urbanization. Each site includes two release locations (150 m apart), randomly assigned to ~~the individuals~~each individual. The peri-urban site is located on the western end of the island of Montreal at the McGill Bird Observatory (MBO; 45˚25'51.4"N 73˚56'18.5"W). The MBO is part of the 30 km^2 large urban park named *Grand parc de l'Ouest*. The suburban site is located in the Town of Baie d'Urfé (45˚24'39.2"N 73˚55'50.0"W). The urban study site is located in the Notre-Dame-de-Grâce neighbourhood of the city of Montreal(45˚27'35.7"N 73˚38'30.2"W). ~~Because cardinals can be found in~~Cardinals are regularly present at all three sites, ~~we assumed~~supporting our assumption that ~~the~~ translocated birds ~~would~~should be able to survive ~~and thrive~~the winter at each location.



Peri-urban site
196 people/km² (SE ± 15 people/km²)

Suburban site
918 people/km² (SE ± 12 people/km²)

Urban site
5361 people/km² (SE ± 83 people/km²)

217

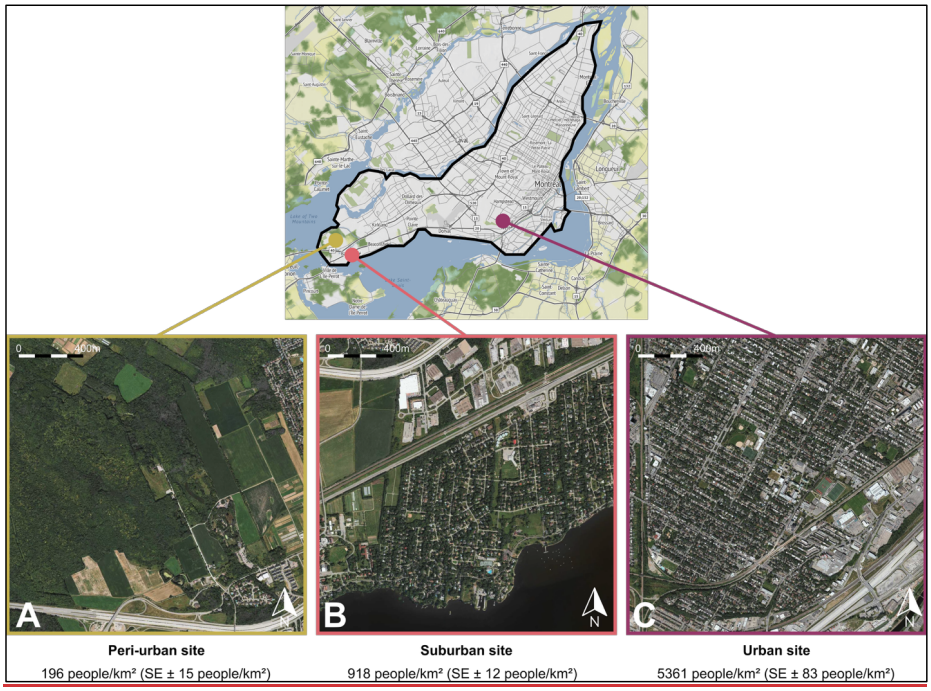


Figure 1. Study locations on the Island of Montreal A) The peri-urban site is at the McGill Bird Observatory (2% of the area covered by streets and/or roads). B) The suburban site (5% of the area covered by streets and/or roads) is located in the residential Town of Baie d'Urfé. C) The urban site (13% of the area covered by streets and/or roads) is located in the Notre-Dame-de-Grâce neighbourhood.

The three translocation sites differed significantly in their level of urbanization, ~~following a Kruskal-Wallis test and a Bonferroni-corrected pairwise Wilcoxon test ($p < 0.001$)~~ based on human population density data from the Cancensus Program R package (von Bergmann et al. 2021), using census data from the Government of Canada for the year 2021. (Kruskal-Wallis test and a Bonferroni-corrected pairwise Wilcoxon test, $p < 0.001$). All three sites are characterized

by relatively high levels of green space compared ~~to the broader urban context of Montreal. The gradient used here ranges from the higher percentages of canopy covers in the peri-urban site~~ to the median canopy cover for the Montreal area (~ 26%) (HealthyDesign.City 2021). As such, ~~this~~our gradient does not fully capture the extent of urbanization present across the entire city and should be interpreted as a localized urbanization gradient. We ~~did try translocating~~initially translocated a few cardinals to ~~an~~a highly urban area with lower canopy cover ~~early on~~in the city core, but due to two immediate mortalities following translocation, we decided to omit ~~this site~~that extreme of the urban gradient from our study.

Radio tagging, banding & translocation

All birds were caught during routine fall migration monitoring at MBO and were fitted with colour bands and Motus NanoTags following standard banding and morphological measurements (see Supplemental Information). Cardinals translocated to sites away from the peri-urban location (suburban and urban), as well as control birds held at the peri-urban site, underwent a holding time of 45 minutes before release to standardize treatment amongst individuals. For the winter of ~~the year~~ 2022-2023 (winter 1), we were able to consistently track 20 birds released as ~~control~~controls at the peri-urban site, which served as a pilot winter to assess the feasibility of the project. During the winter of 2023-2024 (winter 2), we tagged and banded 45 birds, with 15 individuals translocated to the urban site, 15 translocated to the suburban site, and 15 released as controls at the peri-urban site. In 2024-2025 (winter 3), we tagged and banded 42 birds, with 5 individuals released as controls at the peri-urban site, 16 individuals translocated to the suburban site, and 21 individuals translocated to the urban site. ~~During~~

Although tags used in this study were lighter than those used in previous work on cardinals, we considered the potential for tag effects when interpreting our results. Barron et al. (2013) showed

that during the reproductive season, cardinals with tags (weighing 1.6 g [double the 0.8 g of the tags used in the present study]) ~~attached defended their nests less vigorously, but did not change their provisioning effects (Barron et al. 2013).~~defended their nests less vigorously, but did not change their provisioning effects. Cardinal nests with tagged parents also had lower daily survival rates and were less likely to fledge offspring (Barron et al. 2013). However, these tag effects were relatively modest and ~~observed~~occurred across diverse habitats, making it ~~improbable~~unlikely that they would ~~only~~ manifest only in our dense urban site while remaining undetected in peri-urban and suburban locations. Birds were banded under Animal Use Protocol 2007-5447 from McGill University and federal banding permits 10973, 10973B, 10973D issued by the Canadian Wildlife Service, and radio-tagged under Animal Use Protocol 22BF01 and 23BF01 from Environment and Climate Change Canada.

~~Cardinals'~~Cardinal abundance and seasonal distribution

We used the number of cardinals recorded per party~~-~~hour in Audubon Christmas Bird Count data between 1900 and 2015 to compare ~~the~~ historical cardinal abundance ~~of cardinals~~ in Quebec, Canada, ~~against the states of~~with Maine, New York, New Hampshire, and Vermont (United States) (National Audubon Society 2020).

Non-breeding season tracking and home range estimations

~~The birds were~~We monitored tagged birds throughout the non-breeding season using an array of automated Motus towers and handheld telemetry receivers. We conducted manual radio tracking of each individual two to three times per week between the months of October and May each year. The areas monitored were ~5 km^2 (500 ha) at the peri-urban site, ~6 km^2 (600 ha) for the suburban site, and ~18 km^2 (1800 ha) for the urban site. Tracking effort and spatial coverage differed among sites ~~due to logistical constraints associated with~~because weather conditions and

~~the~~landscape structure ~~of the landscape.~~imposed logistical constraints. At the peri-urban site, the ~~absence~~lack of road networks and lower ~~levels of~~ signal interference allowed ~~for~~ the tracking team to move more freely and locate ~~the~~ individuals more ~~rapidly~~quickly, resulting in a smaller monitored area. In contrast, tracking the birds in suburban and urban areas meant that the tracking team's movements were constrained by road access, built infrastructure, private land, and higher levels of radio interference. In addition, individuals released at the suburban and urban sites often moved farther away from their release location, requiring the tracking team to search across larger areas to maintain contact with tagged birds. These differences in spatial coverage reflect practical limitations of tracking birds in the heterogeneous landscape of urban environments and should be considered when comparing movement patterns across sites. The people tracking the birds by hand (trackers) aimed to record at least one GPS location for each individual per week using a Garmin ETREX 22x GPS. Trackers ~~would~~ typically ~~spend~~spent 5 hours tracking per session, twice per week, and started by first ~~tracking down~~searching for newly relocated birds and/or birds with known recent locations. Once these individuals ~~are~~were found, trackers ~~would move~~moved on to expand their search areas to ~~try and~~ find missing individuals. Details on the tracking method ~~used~~ and estimations of precision can be found in the Supplemental Information section.

For all three study sites, we calculated the average home range size using the Kernel Density Estimation (KDE). The home range derived from the utilization distribution (UD) represents the minimum area containing a specified proportion of an individual's space use (~~in the present case, the cardinals'~~i.e., the GPS location recorded in the field) with the kernelUD function from the adehabitatHR: Home Range Estimation R package version 0.4.22 (Worton 1989; Calenge 2023, 2024). For example, the 95% home range represents the smallest area in which there is a 95%

probability of finding the individual. We selected the 95% and 50% ~~and 95%~~ contours to investigate, respectively, the core area used within the home range of the birds and the broader home range excluding the most extreme locations (outliers). A minimum of 5 GPS locations were required to adequately run the function to calculate the KDEs, meaning that individuals with fewer points than this threshold were excluded~~. Between 5 and 48 points per individual were used to estimate the KDEs, with the average number of points per bird being 18~~.

~~Scale-of-effect~~

~~We first identified the scale-of-effect (i.e., the spatial extent at which landscape variables have the strongest relationship with a species' response') by extracting variables at four spatial scales (25, 50, 100, 200 meters; Martin & Fahrig, 2012). These spatial scales were chosen because they cover a wide range of possible scale-of-effect that are relevant in urban ecology, considering that habitat heterogeneity is scale-dependent. However, because scale-of-effect analyses identify the spatial extent that best predicts responses to the tested range, these values should be interpreted as indicating generally fine or coarse spatial responses rather than precise ecological thresholds. For each scale we ran generalized linear mixed models (GLMMs) at each study location using all five predictor variables as well as year, site, and individual ID as random effects. We ranked models using Akaike Information Criterion (AICc) values, which take into account small sample sizes (Burnham and Anderson 2002). The best model was identified as having the lowest AICc value. We also calculated the ΔAICc to evaluate competing models following Burnham and Anderson (2002) (models with a ΔAICc < 2 were considered to have substantial support, competing with the best model and models with a ΔAICc > 10 were deemed to have no support, not competing with the best model). We acknowledge that low per-individual sample sizes might reduce the precision of the scale-of-effect estimates, as models fitted on few observations per~~

~~individual could be susceptible individual-level variation driving apparent scale preferences. To try mitigating this, we included the individuals' IDs as random effect, and we interpret scale-of-effect results at the population level rather than an individual level habitat selection.~~

~~Because the four buffer sizes represent nested spatial extents, we expect the variables extracted from these buffers to be positively correlated. We assessed cross-scale correlations for each predictor type using Pearson's "r". Correlations were highest for human population density (r = 0.96-0.99 across all sites), moderate for mean vegetation heights (r = 0.37-0.94) and road length (r = 0.32-0.95), and lowest for SD vegetation height (r = 0.33-0.84), decreasing with increasing scale separation in all cases. Variables from different scales were never included within the same model—each candidate model uses a single buffer size—so cross-scale correlations do not inflate within-model multicollinearity (all VIF < 10). The buffer comparison therefore constitutes a scale-selection exercise rather than a multi-scale additive approach. Once the scale-of-effect was determined for each location, we conducted a resource selection function.~~

Resource Selection Function

We compared 'used' vs. 'available' habitat using a resource selection function (RSF) for cardinals during the non-breeding season ~~and~~ across ~~an urban gradient.~~our three sites. We used environmental and anthropogenic variables extracted from available spatial data ~~as well that~~ in addition to field-collected ~~in the field~~data. This included mean and standard deviation (SD) of vegetation height (1m resolution) from OpenForest (Government and Municipalities of Quebec 2016), distance to the nearest bird feeder (collected through systematic field observations), human population density data (census unit) using the Cancensus R package (von Bergmann et al. 2021), and spatial data on the streets and ~~road~~roads from the island of Montreal (Ville de Montréal 2018). Unused locations were derived at a 1:1 ratio by creating random locations

within 300 meters of the used location, at a random angle. ~~We used the matched case-control design elaborated by Fortin and have applied a 300 meters buffer to reflect the median daily bird movement (Fortin et al. 2005, 2009). Because this is a population-level resource selection function, in which individual IDs are included as random effects and used-available pairs from all individuals are evaluated together, there is no required minimum number of locations per individual, as it would be the case for an individual-level RSF. The average number of true observations for each individual ranges from 5 to 48 with an average number of points of 18. Having a ratio of 1:1, the same numbers apply for the random observations. The data extracted was scaled and centered before running the models.~~ We used the matched case-control design elaborated by Fortin (2005, 2009) and applied a 300 m buffer to reflect the median daily bird movement. Our analytical design (population-level RSF) included individual IDs as random effects and evaluated all used-available pairs from all individuals together. As such, we did not require a minimum number of locations per individual, as we would for an individual-level RSF. All predictors were scaled and centered prior to analyses.

We built our candidate model set to compare the relative strengths of anthropogenic and natural vegetation variables to predict cardinal habitat use. This included four *a priori* hypothesis-based models including, (1) the "Full model" (Used ~ Mean vegetation height + Sd vegetation height + Population density + Road length + Distance to nearest feeder + (1|Winter) + (1|Site) + (1|ID)), including all of the variables, plus the winter (years of sampling), relocation site and bird ID as random effects to account for the additive importance of anthropogenic and natural vegetation variables, (2) the "Vegetation height" model (Used ~ Mean vegetation height + (1|Winter) + (1|Site) + (1|ID)), assuming the vertical height of vegetation was the dominant variable related to habitat use by cardinals, (3) the "Sd vegetation height" model (Used ~ Sd vegetation height +

(1|Winter) + (1|Site) + (1|ID)), assuming that cardinal habitat use is related to vertical vegetation composition, and (4) the "Anthropogenic" model (Used ~ Population density + Road length + Distance to nearest feeders + (1|Winter) + (1|Site) + (1|ID)), which assumed that cardinal habitat use would be related to anthropogenic variable We also included a "Null" model, with random effects only (Used ~ + (1|Winter) + (1|Site) + (1|ID)). We ranked all ~~fiver~~four models~~-~~ according to the corrected Akaike Information Criterion (AICc) following the method previously described by Burnham and Anderson (2002). We used Variance Inflation Factor (VIF) to test for multicollinearity among predictor variables (variables with VIF values > 10 were considered to have problematic collinearity and were removed)~~ among predictor variables~~ and Hosmer-Lemeshow as a goodness-of-fit test (Hosmer and Lemesbow 1980). VIFs ranged between 1.05 and 3.31 across all predictor variables~~.~~; thus, none were removed. All statistical ~~analysis~~analyses were performed using Program R version 4.4.2 (R Core Team 2024).

Scale-of-effect

We used a data-driven approach to identify the pertinent spatial scale at which variables have the strongest relationship with a species' response (i.e., scale-of-effect, Martin & Fahrig, 2012) by extracting variables at four spatial scales (25, 50, 100, 200 meters). We chose these spatial scales as they spanned a range that was small enough to capture finer responses as allowed by the spatial grain of the vegetation data, but also large enough to capture daily movement of tracked birds. For each scale, we ran generalized linear mixed models (GLMMs) separately for each site type (peri-urban, suburban, urban) using all five predictor variables as well as year, release location, and individual ID as random effects. We ranked models using Akaike Information Criterion (AICc) values, which take into account small sample sizes (Burnham and Anderson 2002). The best model was identified as having the lowest AICc value. We also calculated the

ΔAICc to evaluate competing models following Burnham and Anderson (2002) (models with a ΔAICc < 2 were considered to have substantial support, competing with the best model and models with a ΔAICc > 10 were deemed to have no support, not competing with the best model). We acknowledge that low per-individual sample sizes might reduce the precision of the scale-of-effect estimates, as models fitted on few observations per individual could be susceptible to individual-level variation driving apparent scale preferences. To mitigate this, we included individuals' IDs as a random effect and interpret scale-of-effect results at the population level rather than as individual-level habitat selection.

Because the four buffer sizes represent nested spatial extents, we expect the variables extracted from these buffers to be positively correlated. We assessed cross-scale correlations for each predictor type using Pearson's "r". Correlations were highest for human population density (r = 0.96-0.99 across all sites), moderate for mean vegetation heights (r = 0.37-0.94) and road length (r = 0.32-0.95), and lowest for SD vegetation height (r = 0.33-0.84), decreasing with increasing scale separation in all cases. Variables from different scales were never included within the same model—each candidate model uses a single buffer size—so cross-scale correlations do not inflate within-model multicollinearity (all VIF < 10). The buffer comparison therefore constitutes a scale-selection exercise rather than a multi-scale additive approach. Once the scale-of-effect was determined for each location, we conducted a resource selection function at the site-type-specific spatial scale.

Results

Historical and seasonal context of Cardinals in southern Quebec

To place our winter telemetry results in the context of the species' recent expansion into southern Quebec and its seasonal dynamics within the study region, we summarized long-term regional abundance from Christmas Bird Count (CBC) data and local bird banding records. In southern Quebec, cardinals have increased in abundance since the 1970s as documented in the CBC, reflecting the species' well-documented northward range expansion from the northeastern United States into southern Quebec beginning in the 1960s (Figure 2).

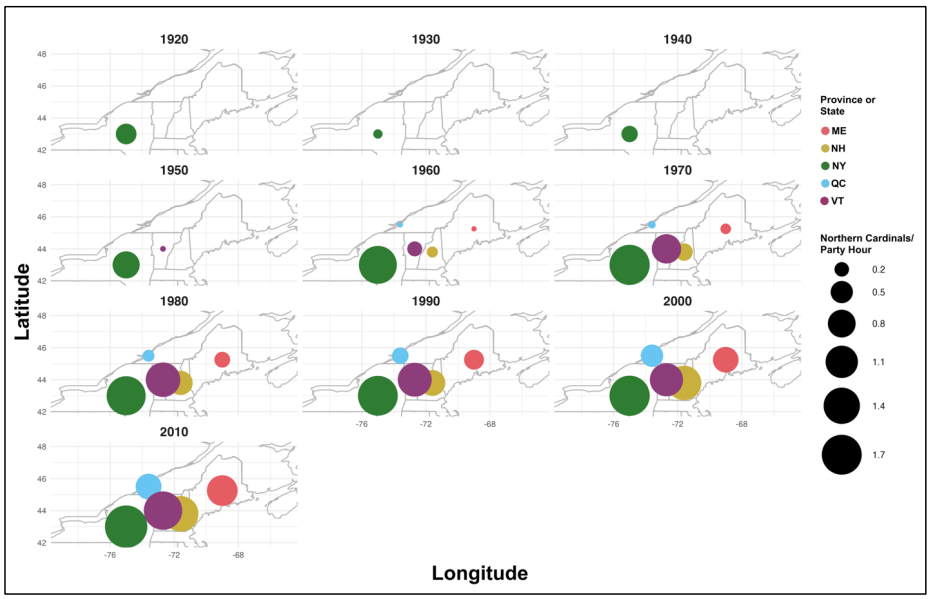


Figure 2. Number of ~~cardinals~~cardinal observations per party hour between the 1920s and the 2010s according to state (United States) or province (Canada) as part of the Audubon Christmas Bird Count~~.~~ (CBC).

Nineteen years of banding data from the McGill Bird Observatory also revealed distinct seasonal patterns in captures (Figure 3). Capture rates were higher during the fall monitoring program (~0.75 birds/h) than during the spring program (~0.27 birds/h), accompanied by a greater proportion of unbanded individuals, indicating an influx of new birds in late fall. Although most fall captures were juveniles or individuals of unknown age (Figure S1B, C), adults also exhibited peaks in mid-May and late July (Figure S1A), suggesting that both post-fledging dispersal and local movements between breeding and wintering areas contribute to seasonal variation in captures.

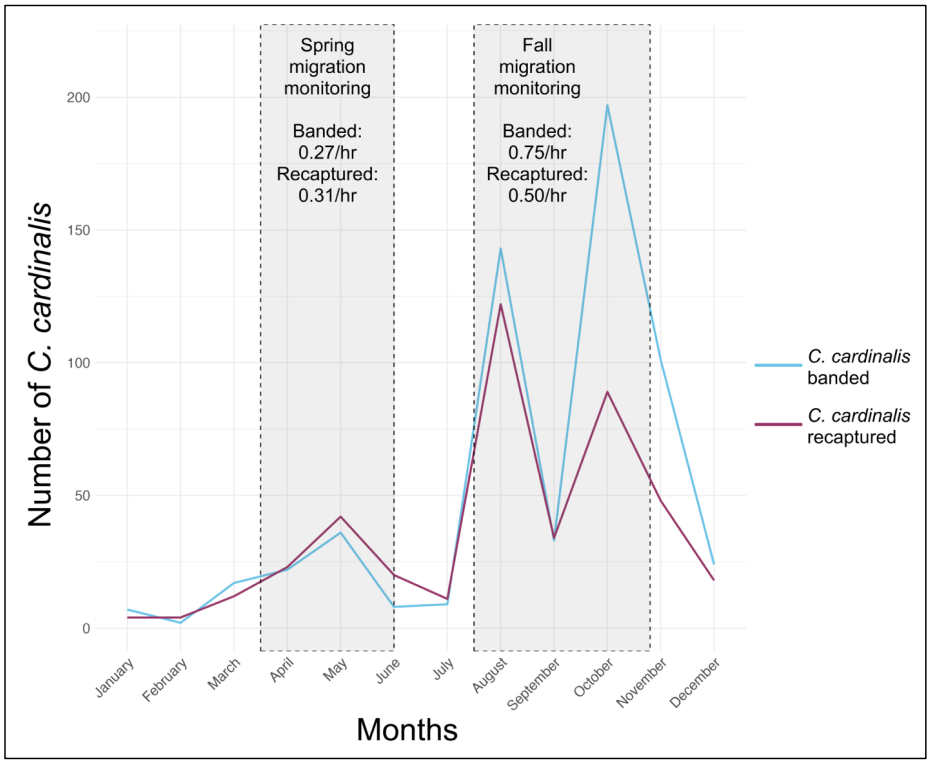


Figure 3. Number of cardinals banded and recaptured at the McGill Bird Observatory (peri-

urban site) between 2004 and 2022 in Montreal, Quebec, Canada. The grey boxes highlight the duration of the spring and fall migration monitoring programs, along with the approximate number of cardinals banded and recaptured per ~~hours~~hour of mist~~-net~~ nets being opened during the season.

Telemetry tracking of cardinals across an urban gradient

~~Across three consecutive winters, we were able to track multiple individuals at all three sites and gather valuable information regarding cardinals' survival through the study (Table 1).~~

Telemetry tracking was conducted 2 to 3 days per week between October and May 2022-2024, for a total of ~288 days of monitoring per winter (Table 1). We recorded a total of 688 unique detections at the peri-urban site, 411 at the suburban site, and 265 at the urban site across the three years. The proportion of cardinals tracked one month post-release differed significantly among sites (Pearson's $\chi^2 = 21.85$, df = 2, $p < 0.001$). Pairwise comparisons with Holm-adjusted p-values showed significant differences between the peri-urban and urban sites ($p < 0.001$) and between the suburban and urban sites ($p = 0.046$), but not between the peri-urban and suburban sites ($p = 0.073$). Similarly, the proportion of cardinals tracked three months post-release differed significantly among sites (Pearson's $\chi^2 = 22.18$, df = 2, $p < 0.001$). Pairwise comparisons showed again significant differences between the peri-urban and urban sites ($p < 0.001$) and between the suburban and urban sites ($p = 0.002$), but not between the peri-urban and suburban sites ($p = 0.518$). The number of cardinals tracked at one and three months post-release, missing individuals, and confirmed mortality events are presented in Table 1.

Table 1. Number of cardinals tracked 1 and 3 months ~~after initial~~ post-release, with apparent survival (48~~.15~~% average) and fated survival (including individuals presumed alive but outside

the study area; 71% average), including known mortality events/probable causes across three sites in Montreal, Quebec, Canada between 2022 and 2025.

Site	Total number of cardinals	# tracked 1 month (apparent survival)	# tracked 3 months (apparent survival)	Missing individuals after 3 months	Confirmed mortality events and probable cause of death
Peri-urban	41	34 (83%)	28 (68%)	13	6 Predation (n = 4) Unknown (n = 2)
Suburban	31	19 (61%)	18 (58%)	13	4 Predation (n = 2) Disease (n = 1) Unknown (n = 1)
Urban	36	11 (31%)	6 (17%)	30	3 Predation (n = 1) Window collision (n = 2)

~~Telemetry tracking was conducted 2 to 3 days per week by three people between October and May 2022-2024, for a total of ~288 days of tracking per winter. We recorded a total of 688 unique detections at the peri-urban site, 411 at the suburban site, and 265 at the urban site. The tracking team covered the maximum amount of ground possible during each outing, to record as many individuals as feasible.~~

Core areas and home ranges of cardinals across an urban gradient

We estimated core areas and home ranges for individuals with ≥ 5 recorded GPS locations (~~35~~36 individuals from the peri-urban site, 13 from the suburban site and 9 from the urban site; Figure 4). ~~Among the 58 individuals~~Between 5 and 48 points per individual were used to estimate the KDEs, with ~~≥ 5 recorded GPS locations, core~~the average number of points per bird being 18. Mean core area sizes (± SE) were 0.15 ± 0.04 km^2 (n = 36) at the peri-urban site, 0.08 ± 0.02 km^2 (n = 13) at the suburban site, and 0.28 ± 0.11 km^2 (n = 9) at the urban site. Mean total home range sizes (± SE) were 0.61 ± 0.13 km^2 (n = 36) at the peri-urban site, 0.56 ± 0.11 km^2 (n = 13) at the suburban site, and 1.53 ± 0.52 km^2 (n = 9) at the urban site. These sample sizes

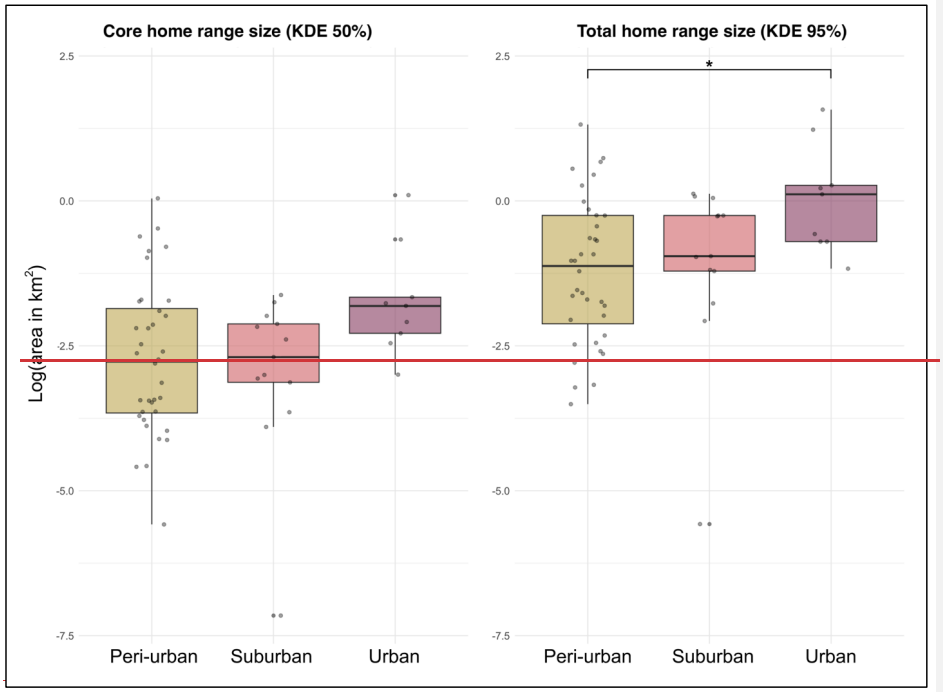
represent the number of cardinals that could be included in the respective KDE analyses. Core and total home range sizes were not normally distributed (Shapiro-Wilk tests: core, $W = 0.624$, $p < 0.001$; total, $W = 0.698$, $p < 0.001$), so areas were log-transformed using a logarithm prior to analysis (Figure 4; Table 2).

Table 2. Number of cardinals that could be included in the analyses along with their respective core areas and home ranges including standard errors across three sites in Montreal, Quebec, Canada between 2022 and 2025.

Site	Total number of cardinals included in the analyses	Core area (km^2)	SE of the core area (km^2)	Home range (km^2)	SE of the home range (km^2)
Peri-urban	36	0.15	0.04	0.61	0.13
Suburban	13	0.08	0.02	0.56	0.11
Urban	9	0.28	0.11	1.53	0.52

After. Following transformation, the data met assumptions of normality (Shapiro-Wilk test: 50% KDE, $W = 0.159970$, $p >= 0.05159$; 95% KDE, $W = 0.150969$, $p >= 0.05150$) and variances were homogeneoushomogeneity of variance (Levene's test: 50% KDE, $F_{2,55} = 1.07$, $p = 0.350$; 95% KDE, $F_{2,55} = 0.61$, $p = 0.547$). There was no significant difference inMean core home range size (95% KDE) was highest at the urban site (Figure 4), but differences among sites, although the effect approached significance were not statistically significant (ANOVA: $F_{2,55} = 2.69$, $p = 0.077$). Post-hoc comparisons show a trend towards the cardinals at our urban site to have larger core home range sizes compared to the suburban and peri-urban sites (Figure 4). There was a significant difference among the three study sites regarding total home ranges (50% KDE; ANOVA: $F_{2,55} = 3.45$, $p = 0.04$). Post-hoc TukeyHSD comparisonsTukey's honestly significant difference tests showed that the only significant difference was betweenthat the total home ranges of cardinals at our urban site beingwere larger than cardinalsthose at the peri-urban site

(mean difference = 1.24, 95% CI [0.09, 2.38], $p = 0.03$~~).~~); no other pairwise differences were statistically significant.


Core home range size (KDE 50%)
Total home range size (KDE 95%)
Log(area in km²)
2.5
0.0
-2.5
-5.0
-7.5
*
Peri-urban
Suburban
Urban

495

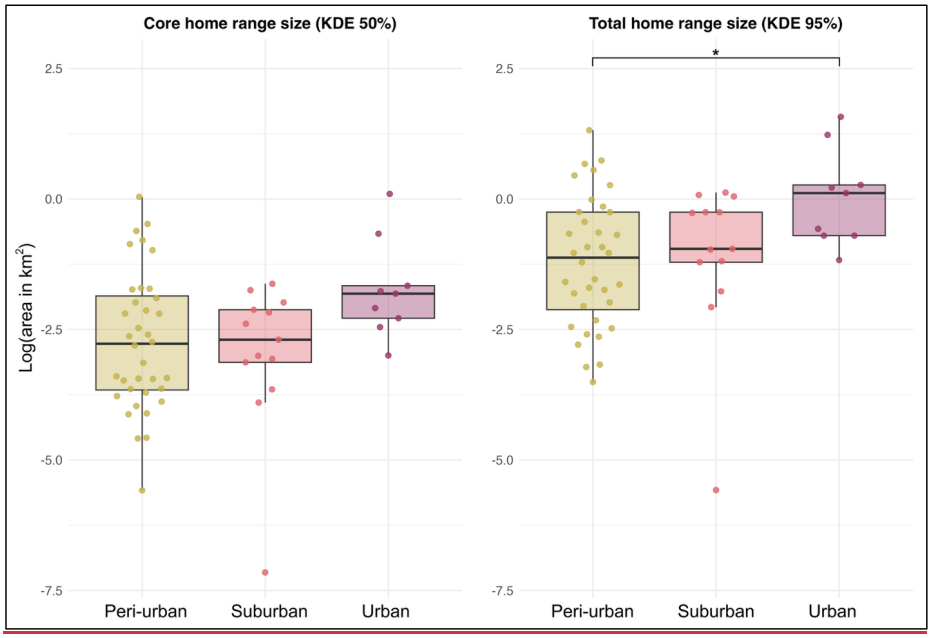


Figure 4. Boxplot comparison of the core (KDE 50%) and total (KDE 95%) home ranges of cardinals in three study sites (peri-urban, suburban and urban) in Montreal, Quebec, Canada during winters between the years 2022 and 2025. ~~The area in km^2 was~~We log ~~-~~transformed~~,~~ the area (km^2), and each data point corresponds to an individual's home range (~~area~~ log ~~-~~transformed area).

Scale of association to habitat

We compared AICc among models fit with landscape covariates measured at four different spatial scales (25, 50, 100, and 200 m radius around resighting locations). At the peri-urban site, the 50 m scale model best explained ~~cardinal's~~cardinal movements (Δ AICc > 4 relative to all other scales). At both the suburban and urban sites, the 25 m scale model performed best (Δ AICc > 10 relative to all other scales). Because the landscape variables we measured at different spatial

scales are inherently correlated, these results indicate the scale at which habitat structure shows the strongest statistical association with ~~cardinal's~~cardinal movements, rather than reflecting a direct causal scale of effect.

Resource selection: used vs. available habitats

Due to differences ~~of~~in urbanization level and scales of effect, resource selection function models were run separately for each study site. For the peri-urban site, the global model explained the most variation seen in the data ~~(> 10 Δ~~(Δ AICc ~~values between~~ > 10 relative to all other models, Supplementary Material Table ~~S1~~S2). The full model reduced deviance by ~11.3% relative to the null model. For the suburban site, ~~the top model was~~we selected the anthropogenic covariates model (human population and road density~~), with~~) as the ~~next best~~most parsimonious model. Although the global model ~~(full model) < 2AICc units from~~had a slightly lower AICc, the ~~top model (Supplementary Material Table S1). The~~ difference between the two models was very small (Δ AICc < 0.5 relative to the top model), indicating similar support for both models. Therefore, we retained the simpler anthropogenic covariates model, which reduced deviance by ~16.6% relative to the null model. Lastly, for the urban site, the ~~full~~global model was the ~~top~~best-supported candidate model~~, but again~~ (Δ AICc > 3 relative to the next best ~~(anthropogenic covariates) was <1 AICc from the top~~ model ~~(Supplementary Material Table S1).~~). The ~~full~~global model reduced deviance by ~10.5% relative to the null model. The best model for each study site was then used to generate prediction plots on the probability of ~~presence~~use of the habitat by cardinals according to the model variables (Table ~~3~~2).

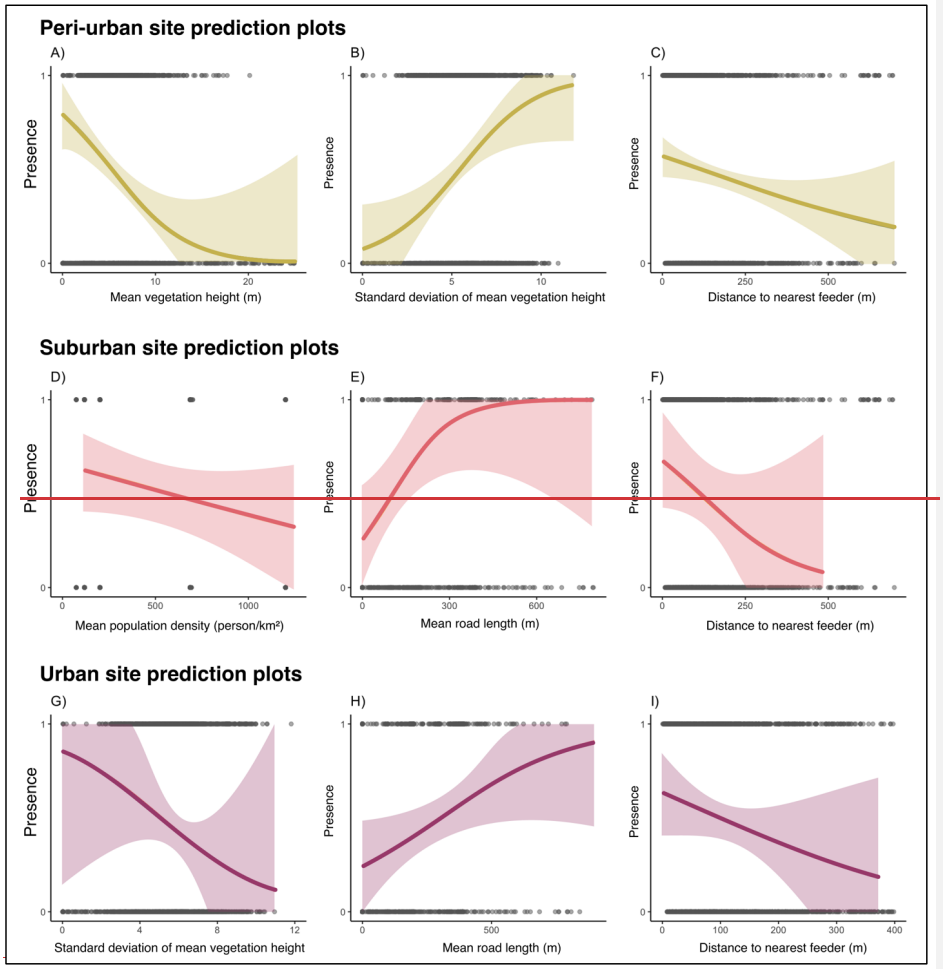
Table ~~3~~2. Model parameters for the three resource selection models predicting the ~~presence/absence~~probability of habitat use of 56 cardinals within (25 or 50 m-radius) areas based on the average vegetation height (in meters) and its standard deviation, along with human-related

environmental features such as human population density (in people/km^2), road length (in meters) and distance to the nearest bird feeder (in meters). The study site, year (winter) of sampling, and the individual's identification (ID) are included as random effects in each model.

Study Site	Variable	Coefficient estimate	Standard error	*z*	*p*
Peri-Urban	*Global model – 50 meters*				
	Mean vegetation height (m)	-1.272	0.153	-8.329	**<0.001**
	Standard deviation of the mean vegetation height	1.107	0.114	9.709	**<0.001**
	Mean population density (person/km^2)	-0.095	0.111	-0.858	0.391
	Road length (m)	0.023	0.110	0.207	0.836
	Distance to the nearest bird feeder (m)	-0.347	0.068	-5.067	**<0.001**
Suburban	*Anthropogenic covariates model – 25 meters*				
	Mean population density (person/km^2)	-0.486	0.135	-3.603	**<0.001**
	Road length (m)	1.238	0.136	9.138	**<0.001**
	Distance to the nearest bird feeder (m)	-0.427	0.128	-3.342	**<0.001**
Urban	*Global model – 25 meters*				
	Mean vegetation height (m)	0.252	0.180	1.399	0.162
	Standard deviation of the mean vegetation height	-0.458	0.191	-2.405	**0.016**
	Mean population density (person/km^2)	-0.212	0.131	-1.620	0.105
	Road length (m)	0.736	0.135	5.438	**<0.001**
	Distance to the nearest bird feeder (m)	-0.392	0.141	-2.772	**0.006**

~~The prediction~~Prediction plots ~~including only~~of the significant fixed effects ~~for~~from the resource selection functions ~~of each locations reveal~~show that ~~the presence of cardinals is associated~~cardinal associations with ~~different~~ urban ~~and~~ habitat features ~~according to the site they spend their winter in~~varied among sites (Figure 5). In the peri-urban site, ~~the probability of presence of the cardinals~~habitat use decreases as the mean vegetation height increases (Figure 5A~~), while~~) and the ~~presence of~~distance to the ~~birds~~nearest bird feeder increases (Figure 5C), and increases with greater variability in vegetation height (Figure 5B). ~~The probability of presence of cardinals also seems to decrease as the distance to the nearest bird feeder increases (Figure 5C).~~ In our suburban site, ~~the probability of presence of cardinals~~cardinals' habitat use shows little relationship with population density (Figure 5D~~),~~) but increases as the mean road length increases (Figure 5E). Like our peri-urban site, the ~~presence~~use of the ~~birds~~habitat by cardinals also decreases as the distance to the nearest feeder increases at the suburban site (Figure 5F). At

the urban site, cardinals' probability of ~~presence~~using the habitat decreases with increasing vegetation height standard deviation, although uncertainty is high (Figure 5G). ~~The presence of cardinals~~The cardinals' association with habitat also increased with mean road length and decreased with increasing distance to the nearest feeders. Across sites, proximity to feeders and structural complexity of the ~~infrastructures~~infrastructure or the vegetation consistently influenced the probability of ~~presence~~habitat use, yet specific relationships varied. Taken together, these results indicate that winter habitat association by cardinals varies across the urbanization gradient. While feeder availability appears to be consistently associated with a higher probability of ~~occurrence~~cardinals using the habitat across all sites, the habitat features associated with cardinal presence differed among the peri-urban, suburban and urban sites. This suggests that patterns of habitat use are largely context~~-~~dependent.


Peri-urban site prediction plots
A)
B)
C)
Presence
Mean vegetation height (m)
Standard deviation of mean vegetation height
Distance to nearest feeder (m)
Suburban site prediction plots
D)
E)
F)
Mean population density (person/km²)
Mean road length (m)
Distance to nearest feeder (m)
Urban site prediction plots
G)
H)
I)
Standard deviation of mean vegetation height
Mean road length (m)
Distance to nearest feeder (m)

558

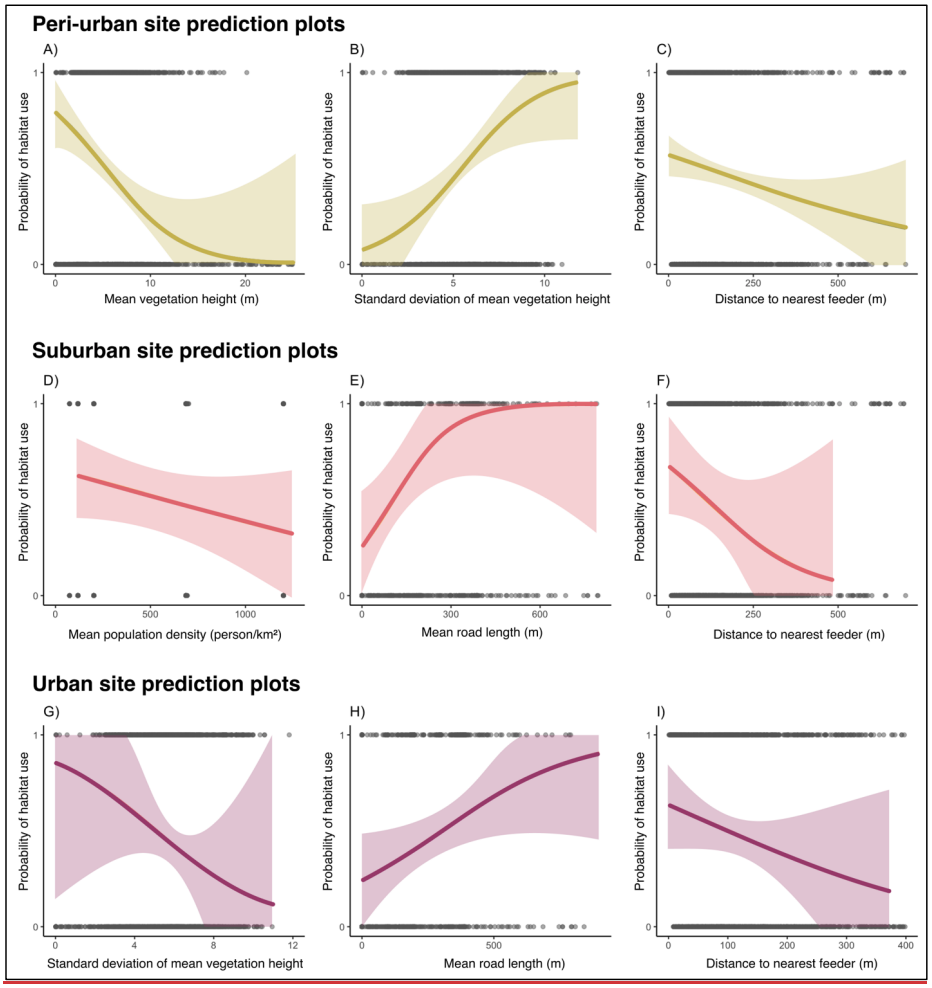


Figure 5. Probability of ~~presence of~~habitat use by cardinals in a ~~given~~50 m radius ~~area~~for the suburban site and 25 m radius for both the peri-urban and urban sites, as predicted by a generalized linear mixed effects model. Shaded areas represent 67% confidence intervals around the predicted probabilities. The model's fixed effect variables are described in Table ~~2~~3, and for each model, random effects were sampling years, translocation site, and bird ID.

Discussion

~~This~~Our study examined winter space use, habitat associations, and apparent survival of cardinals across an urban gradient. Although habitat associations differed among sites, broad patterns emerged. Individuals relocated ~~at~~to the urban site exhibited larger home ranges and lower apparent survival~~,~~ than their counterparts at the suburban and peri-urban sites, while associations with environmental variables varied across the urbanization gradient. Proximity to bird feeders was consistently associated with cardinal occurrence across all sites, and individuals at the suburban and urban sites were primarily resighted in private urban green spaces (UGS) rather than larger public UGS. These findings suggest that winter habitat use by cardinals may vary according to local habitat conditions and resource availability, while anthropogenic food sources appear to be associated with space use across our range of urban environments. Specifically, bird feeders allowed the cardinal range to extend north over the past 80 years in cities, created local movements from UGS to bird feeders each fall, and meant winter habitat use was tied to bird feeder density. More broadly, our results indicate that private UGS may represent an important component of ~~the~~ habitat usage ~~of the~~by cardinals during ~~the non-breeding season~~winter in southern Quebec.

Larger home ranges at urban site and the importance of bird feeders

~~Individuals~~We found cardinals at ~~the~~our most urban site had the largest total (KDE 95%) home ranges ~~(~~, although this finding should be interpreted cautiously given the small sample size at this location (n = 9) and the limited number of locations available for some of the home range estimates. In addition, we did not have a truly rural site, which may also have ~~also~~ had ~~a~~ ~~large~~larger home ~~range).~~ranges. Because we ~~couldn't~~could not capture the full spectrum of

588 urbanization, the relationship to territory size may not be linear (i.e., larger home ranges in rural
589 and urban areas compared to smaller home ranges in peri-urban and suburban areas). *P.*
590 *domesticus* in urban landscapes occupied smaller home ranges compared to their rural
591 counterparts through the breeding season in Belgium (Vangestel et al. 2010). O'Donnell (2020)
592 also suggested that birds in urban environments generally tend to have smaller home ranges.
593 However, female cardinals venture out of their territory more often when they ~~had~~have
594 supplemental feeding, compared to females that ~~didn't~~did not have access to supplemental food,
595 leading to a trend towards larger home ranges (Humbird and Neudorf 2008). In our case, we
596 found the opposite trend. By mapping the location of visible bird feeders, we obtained a coarse
597 estimate of feeder density at all sites: the peri-urban site had ~13 feeders/km^2, the suburban site
598 had ~35 feeders/km^2~~.~~ and the urban site had ~12 feeders/km^2. Although these estimates provide
599 useful context for interpreting site-level differences, they are not representative of all the feeders
600 available and adequately maintained for the cardinals and should therefore be interpreted
601 cautiously. One possible explanation is that the lower feeder density at the urban site, coupled
602 with a potentially more challenging urban matrix, may have been associated with more extensive
603 movements between reliable resource points. Because feeder density was not directly
604 incorporated into our home range analyses, ~~this~~these interpretations should be considered as
605 being on the speculative side. Conversely, the higher feeder density observed at the suburban site
606 might have reduced the distance needed to move among reliable food resources. However, we
607 want to point out that our study was not designed to test this mechanism directly. Future studies
608 that ~~are~~ explicitly ~~quantifying~~quantify feeder availability and ~~incorporating~~incorporate it into
609 movement analyses would be valuable to evaluate whether variation in feeder density in urban

areas is associated with winter home range size for birds exploiting this anthropogenic food resource.

Lower apparent survival at urban site

~~While the apparent survival of cardinals was quite similar between the peri-urban site (on average 65% after 3 months) and the suburban site (on average 57% after 3 months), it was much lower at the urban location (on average 16% after 3 months). Our apparent survival rates (which may have been influenced by emigration, which we could not account for) for the peri-urban and suburban locations are consistent with previously established annual survival in Ohio (50.5% ± 10.3 SE to 78.3% ± 7.6 SE) and in Washington D.C. (between 40 to 70%) in the United States (Shustack 2008; Evans et al. 2015). However, the low survival rates that we are seeing in the urban location appear notably lower than rates previously reported for urban habitats, though direct comparisons are complicated by differences in study design, season and the definition given to apparent survival across studies. Cardinals' annual apparent survival was even greater as the amount of impervious surfaces (used as indicator of urbanization) increases from low to moderate in Washington D.C. (Evans et al. 2015). The lower survival in the urban environment is likely attributable to a higher concentration of anthropogenic threats; it was the only site with confirmed window collisions and indeed 67% of confirmed mortality were window collisions. Urban areas present multiple mortality risks for birds, including window collisions, strikes by vehicle and predation by domestic cats (Loss et al. 2014; Ibáñez-Álamo et al. 2020; Lopes et al. 2024). These risks also apply in suburban areas, but to a lesser extent. Specifically, our urban site had a far higher density of windows than our other sites, with many multi-level low-rise buildings (whereas none at other sites) which we hypothesise could have contributed to elevated collision risk. An alternative explanation is that the individuals translocated to the urban site~~

~~survived but moved farther away from their release site and were beyond the area monitored (> 18 km²). We cannot rule out this explanation, and both processes—true mortality and undetected emigration—may have contributed to the low apparent survival at our urban site. Given the confirmed window collisions and the higher density of anthropogenic hazards, we tentatively favour a mortality driven interpretation but acknowledge this remains uncertain. A further hypothesis is that cardinals faced higher risks by moving in a larger area in the urban habitat, a matrix comprised of many threats including domestic cat predation, window collisions, and vehicle strikes in comparison to the suburban and peri-urban areas.~~We found that the proportion of cardinals tracked three months post-release was similar at the peri-urban (68%) and suburban (58%) sites, but differed substantially at the urban site (17%). These differences provide insights on cardinals' persistence within monitored areas, but they cannot be directly attributed to differences in survival. The peri-urban and suburban estimates are broadly consistent with previously reported annual survival rates for cardinals elsewhere (Ohio: 50.5-78.3%, Shustack 2008; Washington, D.C.: 40-70%, Evans et al. 2015), although a direct comparison is limited by differences in study design and the nature of the data collected. The lower tracking rates experienced at the urban site may reflect an increased mortality associated with anthropogenic hazards. The urban site was the only location where window collisions were confirmed, and these accounted for 67% (2/3) of documented mortalities. Urban environments also expose birds to additional risks, including vehicle strikes and domestic cat predation (Loss et al. 2014b; Ibáñez-Álamo et al. 2020; Lopes et al. 2024). Compared with the other sites, the urban site also contained a substantially higher density of windows, including numerous multi-storey buildings, which may have increased collision risk (Loss et al. 2014a). Alternatively, some individuals may have emigrated beyond the monitored area (>18 km²) and could not conclusively be classified as

dead or alive; their subsequent fates could not be verified in the field. While confirmed mortalities demonstrate that the anthropogenic hazards contributed to losses at the urban site, the available data do not allow us to determine the relative contributions of mortality and emigration to the lower tracking rates we observed.

Small scale of association to habitat regardless of urbanization level

~~Cardinals were associated with habitat features at different spatial scales across our study sites, with the strongest model support occurring at 50 meters for the peri-urban site, and 25 meters for both the suburban and urban sites. While this difference is modest and should be interpreted cautiously, the overall pattern is consistent with cardinals responding to relatively fine-grained habitat features across all sites, regardless of urbanization level. Because telemetry locations and environmental layers contain some level of spatial uncertainty, and because habitat variables measured at adjacent scales are correlated to some level, the estimated scale for the strongest association should be interpreted as an approximation rather than a precise measure of the scale at which cardinals select habitat features. In particular, the spatial error associated with telemetry locations (10-15 m) approaches the difference between our finest scales (25-50 m), further limiting our ability to distinguish between habitat association at these adjacent scales. This aligns with broader research suggesting that the magnitude and direction of species' responses to urbanization reflects landscape-scale differences within cities (Fidino et al. 2021). Moreover, recent work on migratory birds passing through urban areas via stopover sites highlighted the value of multiscale habitat assessment since different processes can come into play at different spatial resolution (Guo et al. 2023). Although our study design did not test this directly, the slightly smaller scale of the strongest model supported for the suburban and urban sites is~~

~~consistent with the idea that habitat configuration may influence habitat association in more urbanized areas. More broadly, our data suggest that habitat associations were detected at a relatively fine spatial scale across the urban gradient, which seems to be consistent with previous work indicating that small habitat patches could contribute and support bird communities in urban areas (Dale 2018). Nineteen years of banding data at the peri-urban site demonstrated that the proportion of unbanded to banded birds was similar over most of the year, implying a mostly resident population, yet there was a distinct peak of new birds in late fall (Fig. S3), suggesting a local, 'very short-distance' migration (including post-fledging dispersal) for cardinals. Our findings highlight the potential value of considering multiple spatial scales when evaluating habitat use in urban areas, and are consistent with previous studies suggesting that fine-scale habitat characteristics may be relevant to resident species in urbanized habitats (Litteral and Shochat 2017; Guo et al. 2023). In our case, bird feeders and vegetation structure had an important impact from the smallest scales (third order habitat selection) to the entire range (first order habitat selection).~~Cardinals were associated with habitat features at relatively fine spatial scales across all study sites, with the strongest model support at 50 m in the peri-urban site and 25 m in the suburban and urban sites. Although this pattern should be interpreted cautiously, given telemetry error (10-15 m) and correlation among habitat variables measured at adjacent scales, it is consistent with cardinals responding to fine-grained habitat features, including bird feeders, across the urban gradient. More generally, our results support previous work suggesting that urban bird responses can vary across spatial scales and that small habitat patches may contribute to avian habitat use in cities (Dale 2018; Fidino et al. 2021; Guo et al. 2023). While our study was not designed to test mechanisms underlying scale dependence, the slightly smaller scale of strongest support in the suburban and urban sites is consistent with the idea that habitat

configuration may become increasingly important in more urbanized landscapes. Nineteen years of banding data from the peri-urban site indicated a largely resident population, as the proportion of banded and unbanded birds remained relatively constant throughout the year. However, a marked increase in newly captured individuals during late fall (Fig. S3) suggests local movements, including post-fledging dispersal and short-distance migration. Across scales, bird feeders and vegetation structure were associated with habitat use from third-order habitat selection to the scale of the entire home range (e.g., first order scale).

Variation of habitat associations across an urbanization gradient

Cardinals were associated with different environmental and anthropogenic habitat variables across an urban gradient, indicating that habitat associations are different among the peri-urban, suburban and urban sites. The one consistent association across study sites was the inverse relationship between ~~cardinals' presence~~habitat use and ~~a shorter~~ distance to the closest bird feeder—which is indeed apparent even at the scale of the entire range (Fig. 2; Robb et al. 2008). This is unsurprising as cardinals are one of the most common species at bird feeders within its range, especially during autumn and winter (Horn et al. 2014). We also found ~~a significant relationship with another anthropogenic variable, wherein~~that cardinals were more likely to be found in areas with higher road cover (which may reflect higher residential development) in both the suburban and urban sites. This pattern is consistent with previous work showing a positive association between cardinals and urbanized landscapes during the breeding season (Evans et al. 2015). Comparatively, cardinal associations with natural cover differed across our urbanization gradient. At the urban site, cardinals avoided ~~the~~ areas of highly contrasted vegetation structures, while at the peri-urban site they were more likely to occupy an area with ~~an elevated~~greater

standard deviation. These findings suggest that vegetation structure may be associated with winter habitat use, which is consistent with a previous study in Delaware and Maryland identifying that vegetation structure is a particularly important variable in urban areas, specifically for cardinals during the breeding season (Duren et al. 2017). Post-fledging cardinals in urban areas strongly select for habitat with a complex understory vegetation (Ausprey and Rodewald 2011). We did not find evidence of the same association with vegetation structure in the peri-urban site. The reason for this discrepancy remains unclear, although variation in habitat availability or habitat heterogeneity among sites may be contributing factors. The importance of vegetation ~~architecture~~structure and composition in birds' habitat selection is most likely scale-dependent (Deppe and Rotenberry 2008). Our scale of effect analysis suggests that cardinals selected ~~their~~ wintering habitats ~~at a larger scale at~~in the peri-urban location~~, compared to~~ at a larger scale than the urban site. It is possible that scale-dependent differences in vegetation cues contribute to this pattern, although we caution that the difference between the scales (25 m vs. 50 m) is modest and that this interpretation remains speculative without a direct test of the vegetation composition effect.

Implications

We highlight the importance of fine-scale habitat features and resource availability in shaping the winter ecology of a resident songbird along an urban gradient. The context-dependent responses we observed underscore the importance of considering local habitat conditions and ~~resources~~resource availability when managing UGS for resident bird species. In particular, the reliance on bird feeders and avoidance of large public UGS suggests that private UGS may play a disproportionately important role in supporting certain overwintering birds. This highlights the often-overlooked importance of private UGS in conservation, even though they are a dominant

and highly diverse part of nature in cities (Goddard et al. 2010; Hutt-Taylor and Ziter 2022). While our study species is not one of conservation concern, it is culturally recognized and may serve as an important species to connect people to nature in urban areas. As urban development intensifies, integrating biodiversity-friendly practices such as preserving vertically complex vegetation and enhancing connectivity between small habitat patches may help sustain populations of resident birds that rely on UGS during the winter. Together, our results highlight how habitat association by a resident bird species is shaped by both landscape-level context and fine-scale resource availability. This multi-scale perspective demonstrates the importance of considering hierarchical habitat use in urban ecology and conservation. More broadly, our findings join a growing body of work emphasizing the ecological importance of urban ~~landscape's~~landscape elements and highlight the need for inclusive UGS planning that includes both public and private green infrastructures.

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Supplemental material

Cardinals tagging procedure

We mist netted cardinals (males and females) at the McGill Bird Observatory (MBO) during October and November of the year 2022, 2023 and 2024. The birds were intercepted using 30 mm Japanese Nylon nets and placed in small cotton bags to reduce stress before handling. As part of the migration monitoring being done at the MBO, we banded all cardinals with a numbered aluminum band issued by the Canadian Bird Banding Office. The individuals were sexed, aged, weighed, and randomly assigned to one of the six possible release sites consisting of translocation sites and control sites at the capture location. In addition to the aluminum band, we fitted all cardinals with a set of heat-sealed coloured plastic bands, each with a unique combination of colours. This allowed visual identification of the individuals without having to resort to a radio tag receiver. Each bird was equipped with a Lotek NanoTag (NTQB-2) weighing 0.8 grams, using soft elastics in a figure 8 harness configuration (the total weight of the tag, harness, and glue was less than 3% of cardinals' body weight). All the radio-tags were registered as part of the Motus Wildlife Tracking System (Frei et al. 2024).

Motus tower locations

During the first winter, only 2 automated receiver towers from the Motus network were deployed. The first tower is located at the peri-urban site (4525'50.4"N 73˚56'18.6"W), while the second tower is located on top of a faculty building at McGill University's Macdonald campus (45˚24'28.3"N 73˚56'20.3"W). For the second winter, those two towers were also used for continuous data location, with the addition of a third temporary receiver tower that we installed at the suburban site (45˚24'39.3"N 73˚55'48.3"W). Finally, during the third winter, a fourth, permanent Motus tower was installed on top of a building at Concordia University's Loyola campus location at the urban site (45˚27'32.6"N 73˚38'30.2"W).

Detailed use of gain and signal strength when tracking

The gain is a value between 0 and 100, representing the amplification of the received signal. By increasing it, the tracker increases how sensitive the receiver is to incoming radio signals. Higher gain values increase the ability to detect weaker signals but are also subject to background noise and interference. Using a lower gain reduces noise but may result in missing weak signals. In urbanized locations, the starting gain will rarely surpass 70-80 as the receiver picks up too much noise and interference. The signal is a value ranging from 0 to 200 for the SRX800 and 0 to 250 for the SRX1200. It represents the strength of the detected transmission from a radio tag, displayed in decibels (dB) on the receiver. A stronger signal indicates that the bird is closer; trackers use this information to estimate distance and direction. Trackers begin their tracking sessions with a gain of 80-70 to increase their chances of detecting nearby individuals. Once the receiver picks up a cardinals' tag, the tracker moves toward the direction with the strongest signal. Once a threshold of 150 dB (SRX800) or 200 dB (SRX1200) is attained, the trackers then lower the gain by 10 and repeat this process until they reach the tagged individual. The trackers only record a GPS point if the signal strength is 150 or 200 dB, with a gain of 40 or less. While visual confirmation was difficult to achieve due to the nature of the release habitats (high density of private property), prior testing with radio tags showed that we could assume the bird to be within 10-15 meters when the gain was ≤ 30 and the signal strength ≥ 150 dB for the SRX800 receivers (or 200 db for the SRX1200). The handheld receivers were either Lotek's SRX800 or SRX1200. They were operated by setting the frequency to 166.380 MHz (Motus tags' frequency for the Western hemisphere) and using both the gain and the signal strength to get as close as possible to the tagged individuals. We were usually able to detect birds that were within 200 meters of our location, and on some rare occasions were even able to pick up signals of birds

located as far as 800 meters away. We did encounter a significantly higher level of static noise when tracking at the urban location, yet it was not enough to prevent us from finding birds within the detection range. Another factor that could have complicated tracking efforts was the occasional aggregation of multiple birds, which can make it difficult to obtain precise location data. In most cases, no more than two birds were observed together. When larger groups were detected, the majority typically dispersed before we could approach close enough to record a GPS location. Overall, however, such aggregations were infrequent and are therefore unlikely to have significantly affected the accuracy of our tracking data.

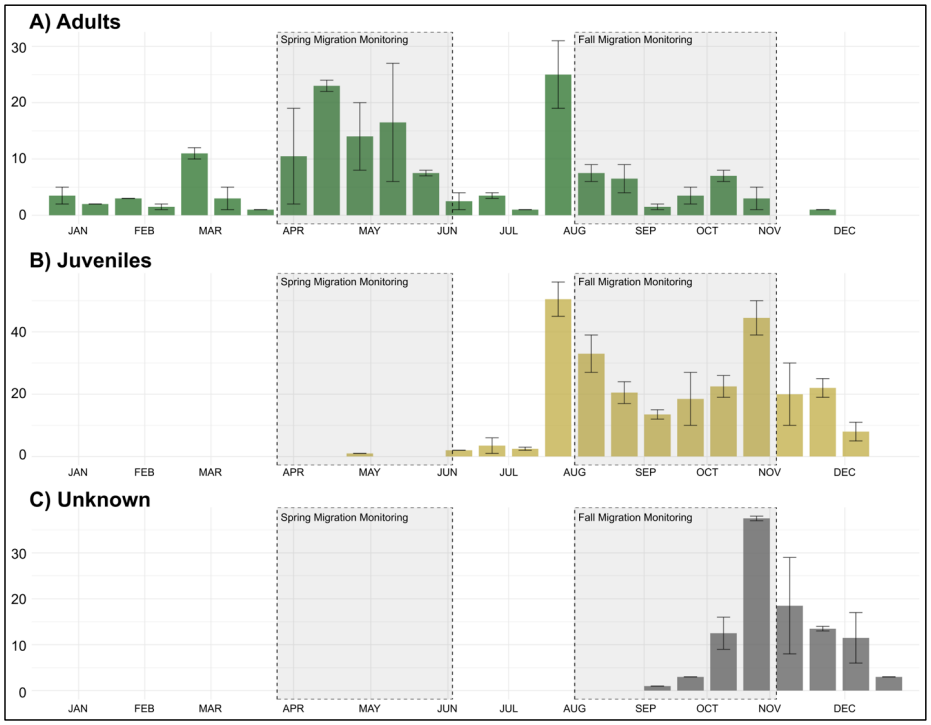


Figure S1. Mean number of cardinals captured at the McGill Bird Observatory (peri-urban site)

between 2004 and 2022 in Montreal, Quebec, Canada separated by age where A) shows adult birds, B) shows juvenile birds and C) shows unknown individuals.

Table S1. Model ranking using AICc of generalized linear mixed models (GLM) with a comparing scale-of-effect of landscape variables for resighting locations (n= 57) of cardinals tracked across three sites in Montreal, Quebec, Canada. The landscape variables include the mean vegetation height, the standard deviation of the vegetation height, the mean human population density, the road length and the distance to the nearest feeder.

Study site	Model	Variables	LL	ΔAICc	Wt
Peri-urban	25 meters	Mean vegetation height + sd vegetation height + mean population density + road length + distance to nearest feeder	-722.48	17.02	0.00
	50 meters		-713.97	**0.00**	**1.00**
	100 meters		-735.69	53.25	0.00
	200 meters		-740.60	46.96	0.00
Suburban	25 meters	Mean vegetation height + sd vegetation height + mean population density + road length + distance to nearest feeder	-296.05	**0.00**	**1.00**
	50 meters		-338.31	84.52	0.00
	100 meters		-344.25	96.39	0.00
	200 meters		-349.92	107.73	0.00
Urban	25 meters	Mean vegetation	-211.09	**0.00**	**1.00**
			-225.64	29.11	0.00

Deleted Cells

50 meters	~~height + sd vegetation height + mean population density + road length + distance to nearest feeder~~	-224.02	25.85	0.00
100 meters		-228.79	35.40	0.00
200 meters				

Table S2. AIC values used to determine the best fitting model to explain cardinal's habitat associations.

Study site	Model	Variables	AICc
Peri-urban	Full	Mean vegetation height + sd vegetation height + mean population density + road length + distance to nearest feeder + (1|Winter) + (1|ID) + (1|Site)	**1443.696**
	Vegetation height	Mean vegetation height+ (1|Winter) + (1|ID) + (1|Site)	1587.124
	Vegetation sd	Sd vegetation height+ (1|Winter) + (1|ID) + (1|Site)	1596.077
	Human	Mean population density + road length + distance to nearest feeder+ (1|Winter) + (1|ID) + (1|Site)	1588.262
	Null	(1|Winter) + (1|ID) + (1|Site)	1616.101
Suburban	Full	Mean vegetation height + sd vegetation height + mean population density + road length + distance to nearest feeder + (1|Winter) + (1|ID) + (1|Site)	**631.7501**
	Vegetation height	Mean vegetation height+ (1|Winter) + (1|ID) + (1|Site)	728.0640
	Vegetation sd	Sd vegetation height+ (1|Winter) + (1|ID) + (1|Site)	724.0187
	Human	Mean population density + road length + distance to nearest feeder+ (1|Winter) + (1|ID) + (1|Site)	**632.1451**
	Null	(1|Winter) + (1|ID) + (1|Site)	726.1005
Urban	Full	Mean vegetation height + sd vegetation height + mean population density + road length + distance to nearest feeder + (1|Winter) + (1|ID) + (1|Site)	**444.7050**
	Vegetation height	Mean vegetation height+ (1|Winter) + (1|ID) + (1|Site)	486.6402
	Vegetation sd	Sd vegetation height+ (1|Winter) + (1|ID) + (1|Site)	485.2496
	Human	Mean population density + road length + distance to nearest feeder+ (1|Winter) + (1|ID) + (1|Site)	447.8691
	Null	(1|Winter) + (1|ID) + (1|Site)	484.8853